Aya Mohamed

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EDUCATION

University of California, Irvine: Henry Samueli School of Engineering ***Mar. 2025 - May 2027***
M.S. Mechanical & Aerospace Engineering – Fluid Dynamics & Propulsion GPA: 3.9/4.00
Zewail City of Science and Technology ***Sep. 2020 - Dec. 2024***
B.S. Aerospace Engineering,with Minor in Electrical engineering Grade: Distinction with Honors • Ranked 2nd

TECHNICAL SKILLS

CAD & Simulations: SolidWorks, Autodesk Inventor, AutoCAD, Ansys Fluent, COMSOL Multiphysics, Proteus
Software & Embedded Systems:Python, MATLAB & Simulink, C/C++, Arduino IDE, LaTeX, Colaboratory
Hardware/Soft Skills: 3D printing, Laser Cutting, Drill Press, Fabrication/Assemblage, Soldering, Technical Writing, Research
AWARDS: ***1st Place IEEE Mobility Challenge (KAUST):*** Jeddah, KSA- 2023|***1st Place Intel ISEF Regional and Finalist:*** Pennsylvania, USA - 2017|***2nd Place - Everybody Can, Organized by US Embassy:*** Cairo, Egypt - 202|***3rd Place Eurocontrol Challenge:*** Toulouse, France - 2024

PROFESSIONAL EXPERIENCES

Aerospace Manufacturing Engineer Intern| Atlas Specialty Products (Novaria Group) | *Anaheim, CA* ***Jul.2026- Sep.2026***
- Assisted in developing and updating work instructions, engineering documentation, and engineering change orders (ECOs).
- Contributed to tooling, fixturing, and workflow improvements while applying Lean Manufacturing principles and aerospace quality standards (AS9100).

Teaching Assistant & Reader| UC Irvine| *Irvine, CA* ***Apr. 2025 - Present***
- TA for Professional Engineering Fundamentals (MAE 202P) and Control Systems (ENGRMAE 170): prepared model solutions, graded assignments, conducted office hours, and led discussion sessions.
- Reader for Mechanics of Structures (ENGRMAE 150) and Dynamics & Control of Aerospace Vehicles (ENGRMAE 175) : graded quizzes, midterms, homework, and finals ensuring consistent, accurate evaluation.

Prototyping Lab Engineer| UCI Beall Applied Innovation| *Irvine, CA* ***Jul. 2025 - Jan. 2026***
- Optimized prototyping workflows for 6 startups by training teams on lab equipment, cutting iteration time by 25% and prototype turnaround by ~20% under high-demand conditions.
- Operated and maintained laser cutting systems (VLS, Epilog) across 3+ machines, producing 55–65 functional parts per week with consistent quality.
- Managed 3D printing operations across 9+ printers (UltiMaker, Bambu, Formlabs 3+), completing 265+ prints per month while minimizing downtime.
- Fabricated mechanical components using CNC milling, mini lathe, mill, and bandsaw, holding tight dimensional tolerances for functional prototypes.

DESIGN PROJECT EXPERIENCES

Brain-Controlled Assistance Robot| **ASRT**| *Egyptian Patent No. 970/2016* ***Jan. 2015 - Jan. 2024***
- Designed and built an EEG-driven robot to assist paralyzed patients, translating brain signals into motion commands; validated on 10 paraplegia and monoplegia patients.
- Designed the full electrical system for a mine-detecting robot, including sensor integration, power distribution, and autonomous triggering logic.

UAV Obstacle Avoidance System| **Arab Academy for Science & Technology** ***Sep. 2020 - Mar.2022***
- Programmed a Python-based real-time obstacle avoidance algorithm for UAVs, handling both static and dynamic obstacles across multiple test scenarios.
- Gained end-to-end knowledge of UAV aerodynamics, structural design, and manufacturing through the full design-build-test cycle.

Electric Race Car Chassis| **Zewail City**| *EVER Competition* ***Mar. 2022- Nov.2023***
- Designed and manufactured chassis components using FEA-informed structural decisions to meet competition safety and weight requirements
- Coordinated with electrical and mechanical subteams to integrate drivetrain and body components within chassis dimensional constraints.